Project Management & Engineering (PME)
Ma'aden Engineering Manual

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REVISION INDEX

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1. INTRODUCTION

1.1 Scope

This document describes the minimum criteria and provisions governing topographic surveys that must be met by a land survey consultant. This includes but is not limited to field procedures, documentation, data collection, computations, equipment, maps and/or drawings.

This specification shall be used with specific work activities. These activities scope of work shall include the specifics for the site. These items include: defined area, number and location of existing bench marks, new bench marks to be included in the work, the contour interval of level work, additional items specific to the site that need to be included in the work, but not specifically noted in this specification.

1.2 Purpose

The purpose of this Specification is to describe the required standards, accepted procedures and performance criteria to be used by Saudi Arabian Engineering and Survey firms, and its agents when conducting surveys. Survey standards, procedures and performance criteria provide consistency of survey accuracy and reliability of mapping products for their customers.

The information included in this Specification has been developed in conformance with applicable directives, policies and procedures, as well as internationally accepted geodetic surveying standards and practices. This Specification is neither a textbook nor a substitute for surveying knowledge, experience or professional judgment. It is intended to establish minimum surveying standards, provide uniform procedures for implementing survey best practices, assure quality and continuity in collection of survey data, and assure compliance with Saudi and relevant survey performance criteria.

This Specification includes both metric and US customary units, procedures for use of RTK GPS, required coordinate system data, and referenced networks. Unless stated otherwise the following measurement unit conventions used for Ma'aden surveys shall be:

- Length or distance measurements shall be in "**meters"**.
- Area units shall be described in "**square meters"**.
- Angular measurements shall be in "**degrees, minutes, seconds"**.
- Control line & boundary line directions shall be described by "**bearings"**.

2. KEY TERMS AND DEFINITIONS

Contractor	An entity, engaged or approved by the Owner, entrusted with a contract to perform Work or cause Work to be performed.

Employer	Saudi Arabian Mining Company (Ma'aden) and its authorized representatives and respective affiliates.
FGCC	Federal Geodetic Control Committee
FGDC	Federal Geodetic Data Committee
GPS	Global Positioning System
NGS	National Geographic Survey
MSL	Mean Sea Level
SI	International system of units.
Site	Geographical location where Services are to be performed.
UTM	Universal Transverse Mercator
Work	The entire work and Services and all things which the Contractor is or may be required to execute, supply and do under the Contract having regard to the Scope of Project, or to cause to be executed, supplied or done, for the purposes of constructing and Commissioning the Plant.

3. REFERENCED DOCUMENTS

This specification requires compliance with the provisions of the latest adopted revision of the relevant standards or codes of the following authorities and technical organizations, as applicable, and specifications which shall be considered to form part of this specification, where applicable.

3.1 Statutory Regulations

All work performed and materials furnished shall comply with the applicable statutory regulations, codes and other requirements of the Kingdom of Saudi Arabia (including but not limited to directives of the High Commission for Industrial Security, Royal Commission regulations applicable to the Site, regulations of the Presidency of Meteorology and Environment, Saudi Building Code and other applicable codes and regulations). The Contractor shall ensure that he is fully aware and informed of all Government laws, local ordinances, regulations and all orders and decrees of bodies or tribunals having any jurisdiction or authority over the Project and/or Site, which in any manner affect the Project and those engaged or employed on the Project, or which in any way affect the conduct of the Work. The Contractor shall at all times observe and comply with all such Government and/or Ministry laws, bylaws, ordinances, codes, regulations, orders and decrees.

3.2 Industry Codes and Standards

All equipment and systems shall conform to the following:

- Standards and Specifications for Geodetic Control Networks, Federal Geodetic Control Committee & Federal Geodetic Data Committee (FGCC/FGDC).

3.2.1 International Codes and Standards

3.2.1.1 National Society of Professional Surveyors

ALTA/NSPS	Minimum Standard Detail Requirements for ALTA/NSPS Land Title Surveys

3.3 Ma'aden Standards

MD-101-SMEM-EG-GE-PRO-0001	CAD Standards
MD-101-SMEM-EG-GE-PRO-0002	Document and Revision Numbering
MD-101-SMEM-EG-GE-TMP-0001	Standard drawing templates A0

4. FIELD SURVEY INITIATION

4.1 General

The Works, including field surveys, drawings, reports and other deliverables shall be carried out under the signature of a Contractor legally entitled to practice surveying in the country, state, region, province or similar of the Works and meet any specific prequalification requirements of the Employer.

4.1.1 Safety

The Contractor shall take all necessary actions to ensure the safety of personnel throughout the field survey investigation including mobilization and demobilization of all necessary equipment and personnel.

4.1.2 Work Permit

Local governments may require permits to be obtained for the survey of topographic features. The requirements may vary depending on local engineering and environmental ordinances.

It shall be the responsibility of the Contractor to identify the relevant governing bodies pertaining to the requirements of conducting topographical survey work. All required permits shall be obtained by the Contractor before the commencement of the field survey works, with adequate time provisions in place for the approval of required permits by the responsible government agencies.

4.1.3 Deliverables

Refer to MD-101-SMEM-EG-GE-PRO-0001, MD-101-SMEM-EG-GE-PRO-0002, and MD-101-SMEM-EG-GE-TMP-0001

5. QUALITY CONTROL

5.1 Horizontal Control

The accuracy of horizontal measurements for establishing primary horizontal control shall meet the requirements of a third-order, class I survey, as defined by FGCC Standards (1984), Table 2.1, unless specified otherwise. Secondary horizontal control networks shall be closed and traverses shall be tied to primary horizontal controls. The accuracy of controls shall meet the requirements of a third-order, class II survey, as defined by FGCC Standards (1984), Section 2.1, unless specified otherwise.

5.2 Vertical Control

The accuracy of vertical measurements to establish primary vertical control shall meet the requirements of a second-order, class II survey, as defined by FGCC Standards (1984), Table 2.2, unless specified otherwise.

Secondary vertical control networks shall be level loops tied to primary vertical controls. The accuracy of controls shall meet the requirements of a third-order, class II survey, as defined by FGCC Standards (1984), Section 2.2, unless specified otherwise.

5.3 Tolerances

The Contractor shall meet the following tolerances when making both vertical and horizontal measurements unless otherwise specified in other Standards/Specifications:

- ± 30 mm for ground elevations
- ± 75 mm for surface contours
- ± 6 mm for measurements on concrete, and
- ± 3 mm for tie-in work and survey of existing facilities.

5.4 Units

All measurements shall be in SI Units.

Length shall be measured in meters (m) correct to two decimal places. Areas shall be worked out in square meters (sqm) and volume in cubic meters(cum) rounded off to two decimal places. Elevations shall be measured in meters (m) correct to three decimal places for monuments and bench marks and two decimal places for setting out points of the Work.

6. EXECUTION

The Contractor shall be responsible for the proper execution of the Work to meet the requirements of the Specification, drawings or as directed by the Employer.

The Work at Site shall be carried out under the full time supervision by a qualified Engineer or a senior Surveyor. The Engineer or senior Surveyor shall be responsible for and capable of coordinating the work of the surveying teams, setting out the work accurately, identifying the type of instruments to be deployed and methodology of

surveying to achieve the requirements of the Specification.

Appropriate instruments should be deployed to achieve the specified accuracy of the work. Proper precautions for avoiding graduation errors, instrumental and personal errors should be scrupulously observed.

6.1 Scope

Accurately show all significant natural and manmade features within the limits of mapping (with Northing and Easting coordinates and Elevation), to include, but not limited to, the following:

- Edge of pavement or other surface treatment (gravel, grass, plantings)
- Roads, walks, paved aprons, steps, loading/unloading docks
- Truck Scales
- Railroads including top of rail elevations, points of switch, derailers, etc.
- Pipe Racks, including rack foundations and direction/number of pipes on rack
- Above ground pipes and sleepers, including size and material type of pipe, valve assemblies, etc.
- Culverts, headwalls, tunnels
- Walls, fences, and bollards
- Aerial wire lines, Power/Utility poles, Light Poles and minimum vertical clearances.
- Buildings and other structures (such as transformers, tanks, towers)
- Piers/Foundations extending above grade; equipment pads
- Exposed columns
- Building Corners (foundation and/or face of siding) and finished floor elevation ("ground floor only")
- Storage areas (do not locate movable objects stored in these areas)
- Parking areas; recreation areas
- Tree clumps, and landscaped areas
- Shorelines, streams, ditches, channels, dikes & berms and other drainage ways (include date of water level measurement)
- Manhole covers, drainage inlets, process drains, trench drains, including associated top cover and all invert elevations by cardinal direction
- Storm water drainage and sanitary sewer systems, including pipe size and type of material

- Sanitary sewer lift stations with associated inlet and outlet invert elevations by cardinal direction
- Fire Hydrants, post indicator valves, gate valve assemblies, hose houses
- Gas Meters, gas valves & wells
- Low Voltage lines; closed circuit or cable tv, telephone, security
- Property and Control Monuments
- Water wells
- Ground Water Monitoring Wells

Show demarcation lines lying within the mapping area, and include a complete map legend for the following:

- Property lines
- Set back lines
- Easement lines
- Right-of-way lines
- Government land survey lines
- Associated monuments
- Established and/or Protected Wetlands
- New and existing benchmarks and all reference monuments and sources of information/location used for the survey

All earth deposits, depressions, hills and valleys within the area(s) are to be surveyed and plotted on maps by contours. Necessary leveling work of the entire area(s) are to be surveyed and plotted on maps by establishing horizontal location so that sketching of contours for the area(s) can be done at specified intervals and in specified scales on maps.

Where grid establishment is not possible e.g. in the case of steep slopes and densely vegetated areas , the method of survey, contour intervals etc shall be as directed by the Employer. Any unusual conditions or formations on the ground such as rock outcrops, springs/falls etc shall also be surveyed accordingly to facilitate plotting on the maps.

The field work shall include the following:

- Establishing horizontal and vertical controls
- Locating reference grids and bench marks in the area
- Surveying spot levels and plotting contours, and
- Locating and plotting natural and man made features as described above.

The grid for the survey work shall be established in north-south & east-west direction corresponding to magnetic north or the plant north as directed by the Employer.

Clearance and cutting of trees as required to facilitate the survey work also forms part of the Work. Necessary permission of concerned public bodies shall be secured by the owner. The Contractor shall ensure that the minimum amount of bush-land is cleared and trees are properly cut under the direction of public bodies.

6.1.1 Benchmarks, Reference and Grid Pillars

The Contractor shall set up a new bench mark by fly-leveling from the nearest two approved bench marks or other available source as approved by the Employer.

The Work shall include constructing permanent reference pillars at suitable locations as approved by the Employer. These reference pillars shall be labeled permanently with their respective co-ordinates and reduced levels for future use. The bench mark and reference pillars shall be shown on the survey drawings.

Benchmarks shall be located in strategic locations, protected from traffic and constructed as per the directions of the Employer. Benchmarks shall be clearly engraved with the reduced level and co-ordinates with respect to the survey grid. Benchmarks shall be painted with anticorrosive paint.

7. PROGRESS REPORTING

The Contractor shall prepare and submit a progress report in duplicate every week to the Employer indicating the status of the Work, methodology adopted for surveying, instruments deployed, staff working on the Site and difficulties encountered during execution of the Work etc.

The submission of such reports and review thereof by the Employer shall not be deemed to absolve the Contractor of his responsibility of timely completion of the assignment as per the time schedule indicated or any other responsibility.

8. SUBMISSION OF DOCUMENTATION

The Contractor shall submit hardcopies and softcopies of all documentation as per the contract, including but not limited to:

- survey report
- survey maps
- field notes
- calculations.

Survey maps shall also be provided in electronic format as agreed by the Employer.

Field data shall be submitted to the Employer from time to time if requested, and as per

progress of the work.

The Contractor shall submit survey maps of the Site in 1:1000 scale, unless otherwise noted or required for legibility, indicating grid lines and contour lines, demarcating all features as listed in Section 6.

Overall plant and facility drawings and maps shall show a minimum of three (3), or as agreed with the Employer, survey monuments with UTM coordinates and MSL elevations.

All documents shall be submitted for review and approval by the Employer.

All documentation submitted shall generally be compatible with the following software systems, or file formats, unless advised otherwise by the Employer:

- Microsoft Office suite
- dtm, .dxf, .dgn, .dwg, .CSV
- pdf.

8.1 Survey Report

The report should give an introduction to the Site, methodology adopted for surveying the areas, calculation of errors, transfer of bench marks and any other calculations required for surveying and preparation of the survey maps. Details of trees with their name, numbers and girths shall also form part of the survey report.

The survey report should also cover the following:

- General Site observation such as location of existing structures, installations, access roads, rivers and irrigation canals etc
- Presence of any wells and/or tube wells in the Site indicating their water level also
- Any rock outcrops on the Site, and
- Existing drainage pattern of the Site, possibility of water logging and high flood level of the area.